Progress Report 3

Week 3: Baselines B1–B3, Proposed Model P, Evaluation Engine, and Beam Search

Cole Rodney Migisha	b29657	Charis Opol	b29654
Ainomugisha Faith	b27518	Lisa Lusinga Namugenyi	b30300
Ansasire Nehum Naamaye	b29785	Wakisa Isabelle	b30311

Project: LipScribe – Lip-Assisted Speech Recognition for Robust Live Captioning 9 November 2026

1 Summary

In Week 3 (and early Week 4 follow-through), the team completed all deliverables for Milestones **M3** and **M4** and finished the full training and evaluation of Proposed **Model P** (**M5**). All three baseline families—audio-only **B1**, visual-only **B2** (v1 & v2), and concatenation fusion **B3** (v1 & v2)—along with all four initialisation variants of **Model P** (bidirectional cross-modal attention + per-frame gated fusion + learned missing-modality embedding $\mathbf{e}_{\text{miss}}$) have been trained on the full GRID corpus on our Alienware RTX 4080 SUPER workstation and evaluated across all five acoustic conditions (clean, 10, 5, 0, −5 dB) on the four held-out test speakers.

The empirical findings validate both our multimodal hypothesis and our architectural choices:

1. **B1 (Audio-Only)** achieves 1.96 % test WER on clean speech (1.76 % with grammar decoding), but degrades nearly five-fold under severe noise (9.45 % greedy, 8.35 % grammar at −5 dB).
2. **B2 (Visual-Only)** is completely invariant to acoustic noise (23.04 % greedy, 21.03 % grammar test WER across all SNRs for v1), providing a noise-immune floor in adverse environments.
3. **B3 (Concatenation Fusion v2)** reduces B1's error at every SNR (**1.16 %** clean, **5.21 %** at −5 dB greedy; **0.77 %** clean, **3.83 %** at −5 dB with grammar decoding), but collapses catastrophically when a modality is missing at test time (25.18 % without video on clean speech; 75.31 % without audio).
4. **Proposed Model P (P4, initialised from B1 + B2 v2)** achieves our best overall greedy test WER (**1.02 %** clean, **2.51 %** mean across SNRs) and best clean grammar WER (**0.71 %**, 0.34 % CER). Crucially, Model P resolves B3's missing-modality fragility via $\mathbf{e}_{\text{miss}}$ and dynamic gating: dropping video yields **1.88 %** clean WER (matching standalone B1's 1.96 %), dropping audio yields **23.78 %**–**31.57 %** WER (matching standalone B2), and the mean fusion gate $\bar{g}$ automatically shifts from 0.804 (clean) down to 0.691 (−5 dB) to rely more heavily on lip movements as acoustic noise intensifies.

1.96 %
B1 audio-only test WER, clean (9.45 % at −5 dB)

23.04 %
B2 visual-only test WER (21.03 % grammar; constant across SNR)

1.02 %
Model P4 greedy test WER, clean (0.71 % grammar; 2.51 % mean)

1.88 %
Model P4 no-visual fallback WER vs. 25.18 % for B3 v2

2 Progress against the Milestones

Milestone	Due	Status and Progress
M1 Environment and dataset verified	Sun 25 Oct	**Done** – closed 29 Oct
M2 Preprocessing and alignment	Sun 1 Nov	**Done** – closed 3 Nov; full corpus extraction complete
M3 & M4 Baselines B1, B2, B3	Sun 8 Nov	**Done** – closed 4 Nov; all baselines (v1 & v2) trained and tested
M5 Multimodal fusion (model P)	Sun 15 Nov	**Done** – cross-attention, gated fusion, $\mathbf{e}_{\text{miss}}$ & 4 full runs complete
M6 & M7 Tuning and evaluation	Sun 22 Nov	**In progress** – 4-way curriculum & modality ablations evaluated
M8 & M9 Application, testing, demo	Sun 29 Nov	Planned

3 Week 3 & Early Week 4 Work Breakdown and Deliverables

Table 1 summarises the status and verified outputs for each Week 3 task and the completed Week 4 fusion milestones.

Table 1: Week 3 and early Week 4 work breakdown cards and verified outcomes.

Card	Task	Owner	Status	Output and Evidence
W3-01	Complete B1 (audio)	Charis	**Done**	52 epochs (best ep 43); test WER 1.96 % (clean) to 9.45 % (−5 dB)
W3-02	Evaluation script	Lisa	**Done**	`scripts/evaluate.py`: per-speaker metrics, hypotheses CSV, gate & ablation hooks
W3-03	Visual encoder V1	Cole	**Done**	3D-CNN + 2-layer BiLSTM (5.25 M params); V2 ResNet-18 front end (15.0 M params)
W3-04	Visual augmentation	Faith	**Done**	Crop jitter, horizontal flip, scale jitter (±10 %), spatial cutout, time masking
W3-05	Train B2 (visual-only)	Cole	**Done**	B2 v1 (28 ep): 23.04 % WER; B2 v2 (13 ep, 50m budget): 32.20 % WER
W3-06	Implement & train B3	Faith	**Done**	B3 v1 (22 ep): 2.41 %–7.35 % WER; B3 v2 (7 ep): 1.16 %–5.21 % WER
W3-07	CTC beam & grammar	Isabelle	**Done**	Prefix beam search (17.3 ms/utt) & GRID Viterbi grammar decoder (11.0 ms/utt)
W3-08	Exp X5: audio features	Lisa	Planned	Deferred to Week 5 to prioritise baseline completion and Model P fusion
W3-09	Experiment tracking	Nehum	**Done**	`results.csv`, per-run `history.csv`, `plot_all_model_comparisons.py`
W3-10	Gradio Mode A skeleton	Isabelle	**In progress**	Preprocessing and video playback wired; upgrading to Model P4 checkpoint
W4-01–05	Proposed Model P & curriculum	Cole / Charis	**Done**	`fusion_attention.py`, 8-utt overfit (CER = 0), 4 full-corpus Model P runs (`P1--P4`)

4 Official Held-Out Test Results Across All Models

Table 2 presents the official held-out test evaluation across all five noise conditions on the 3,998 utterances from unseen test speakers `s4`, `s9`, `s14` and `s24` (validation speakers: `s11`, `s19`, `s26`). Figure 1 illustrates the test WER trajectories across SNR levels.

5 Why v2 Used ResNet-18, Modality Ablations, and Grammar Decoding

Table 2: Official Held-Out Test Evaluation: Word Error Rate (WER %) and Character Error Rate [CER %] with Greedy CTC Decoding across Noise Conditions (3,998 utterances per SNR).

Model Configuration	Clean	10 dB	5 dB	0 dB	−5 dB	Mean WER
B1 Audio-Only (52 ep)	1.96 [0.8]	3.02 [1.2]	3.79 [1.5]	5.44 [2.4]	9.45 [4.7]	4.73 %
B2 Visual-Only v1 (`Conv3D`, 28 ep)	23.04 [14.0]	23.04 [14.0]	23.04 [14.0]	23.04 [14.0]	23.04 [14.0]	23.04 %
B2 Visual-Only v2 (`ResNet18`, 13 ep)	32.20 [18.6]	32.20 [18.6]	32.20 [18.6]	32.20 [18.6]	32.20 [18.6]	32.20 %
B3 Concat Fusion v1 (Scratch, 22 ep)	2.41 [1.1]	2.88 [1.4]	3.36 [1.6]	4.29 [2.2]	7.35 [3.8]	4.06 %
B3 Concat Fusion v2 (`B1+B2v2`, 7 ep)	1.16 [0.5]	1.66 [0.7]	1.98 [0.8]	2.81 [1.3]	**5.21 [2.6]**	2.56 %
Model P1 (Scratch `Conv3D`, 11 ep)	12.58 [6.4]	13.05 [6.6]	13.55 [7.0]	14.49 [7.6]	16.28 [8.6]	13.99 %
Model P2 (Init `B2v1` only, 11 ep)	18.19 [10.7]	18.20 [10.7]	18.53 [10.9]	18.93 [11.2]	19.85 [11.8]	18.74 %
Model P3 (Init `B1+B2v1 Conv3D`, 10 ep)	2.13 [0.8]	2.67 [1.0]	3.27 [1.2]	4.33 [1.8]	7.30 [3.4]	3.94 %
Model P4 (Init `B1+B2v2 ResNet18`, 6 ep)	**1.02 [0.4]**	**1.53 [0.6]**	**1.87 [0.8]**	**2.79 [1.2]**	5.34 [2.5]	**2.51 %**

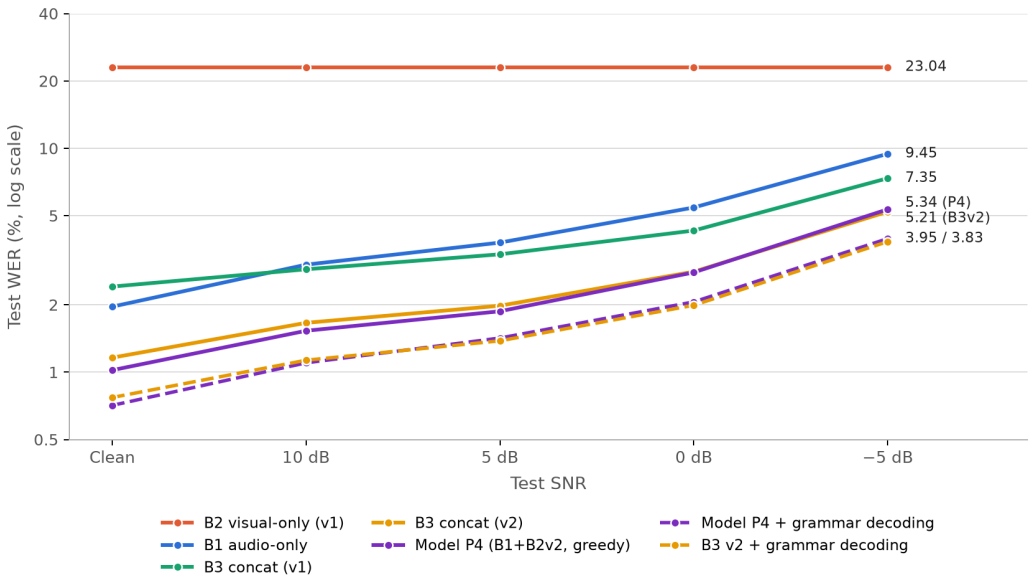


Figure 1: Test WER across acoustic noise levels (log scale; greedy decoding unless stated). B1 degrades steeply as SNR falls and B2 is constant. Both B3 v2 and Proposed Model P4 outperform B1 at every SNR, with P4 achieving the lowest clean error (1.02 % greedy, 0.71 % grammar).

5.1 Why ResNet-18 (v2) Outperforms 3D-CNN (v1) in Multimodal Fusion

Although standalone **B2 v2** (`resnet18_bilstm`, capped at a 50-minute / 13-epoch budget with aggressive spatial cutout and time masking) recorded a higher standalone WER (32.20 %) than 28-epoch **B2 v1** (23.04 %), our controlled 30-minute full-corpus comparison between **Model P3** (`B1 + B2 v1 Conv3D`) and **Model P4** (`B1 + B2 v2 ResNet-18`) proves that the `ResNet-18` visual encoder provides substantially superior representations for multimodal fusion:

- **52.1 % lower clean test WER and 38.0 % lower validation WER:** Holding the audio checkpoint (`B1`), fusion module, learning rate, and 30-minute training budget identical, swapping `B2 v1` (`Conv3D`) for `B2 v2` (`ResNet-18`) cuts validation WER from 14.17 % to **8.82 %** and greedy test WER from 2.13 % to **1.02 %** (clean) and from 7.30 % to **5.34 %** (−5 dB).
- **Architectural reason:** `v1` flattens a 96×6×6 = 3,456 feature map after only 3 plain convolutions, making it sensitive to speaker lighting and crop shifts. `v2` combines a 5-frame 3D spatiotemporal stem (`Conv3d` 5×7×7) with per-clip zero-mean/unit-variance normalisation (`input_norm="clip"`), a 16-layer 2D residual trunk with skip connections, and global `AdaptiveAvgPool2d(1)`, producing speaker-invariant viseme features (detailed in our companion note `resnet_vs_conv3d_justification.pdf`).

5.2 Missing-Modality Robustness and Noise-Adaptive Gating (Model P vs. B3 v2)

Table 3 compares **B3 v2** and **Model P** when an entire modality is dropped at test time (`--drop-modality visual` or `--drop-modality audio`) alongside Model P4's mean fusion gate $\bar{g}_t = \sigma(\mathbf{w}^\top[\tilde{\mathbf{a}}_t; \tilde{\mathbf{v}}_t] + b)$.

- **Graceful unimodal fallback:** When video is dropped, **B3 v2** degrades from 1.16 % to 25.18 % WER (clean), whereas **Model P4** substitutes $\mathbf{e}_{\text{miss}}$ and achieves **1.88 %** clean WER—matching standalone B1 (1.96 %). When audio is dropped, B3 v2 collapses to 75.31 %, whereas Model P maintains 23.78 % (P2), 25.65 % (P3), and 31.57 % (P4, beating standalone B2 v2's 32.20 %).
- **Verified noise-adaptive gate shift:** As SNR drops from clean to −5 dB, Model P4's mean gate $\bar{g}$ decreases monotonically from **0.804** to **0.691** ($\Delta\bar{g} = -0.113$, more than 2.2× the shift of P3's `Conv3D` gate from 0.884 → 0.834), shifting up to 30.9 % fusion weight onto the visual stream under heavy noise.

Table 3: Missing-modality test ablation (WER %) and Model P4 mean fusion gate $\bar{g}$ across SNRs.

Model / Test Condition	Clean	10 dB	5 dB	0 dB	−5 dB
Model P4: Audio + video (bimodal greedy)	**1.02**	**1.53**	**1.87**	**2.79**	5.34
Model P4: Audio only (`--drop-modality visual`)	**1.88**	**2.99**	**3.59**	**5.44**	**10.36**
B3 v2: Audio only (`--drop-modality visual`)	25.18	30.69	33.55	38.40	47.41
Model P2 / P3 / P4: Video only (`--drop-modality audio`)	**23.78 % (P2) / 25.65 % (P3) / 31.57 % (P4)**				vs. **75.31 % (B3 v2)**
Model P4 Mean Fusion Gate $\bar{g}$ (audio weight)	**0.804**	**0.786**	**0.769**	**0.740**	**0.691**

5.3 GRID Grammar-Constrained Decoding

Using our exact CTC Viterbi decoder over the 64,000-sentence GRID grammar (`lipscribe/decoding/grammar.py`, 11 ms/utterance on CPU), every model achieves substantial further error reductions without retraining (Table 4). **Model P4 + grammar** reaches **0.71 %** clean WER (0.34 % CER) and **3.95 %** at −5 dB (58.2 % relative reduction over greedy B1).

Table 4: Test WER % comparing greedy vs. GRID grammar-constrained decoding on the same checkpoints.

Model	Clean greedy	Clean grammar	−5 dB greedy	−5 dB grammar	Mean grammar
B1 Audio-Only	1.96	1.76	9.45	8.35	4.26 %
B2 Visual-Only (v1 / v2)	23.04 / 32.20	21.03 / 28.26	23.04 / 32.20	21.03 / 28.26	21.03 / 28.26 %
B3 Concat (v1 / v2)	2.41 / 1.16	1.86 / 0.77	7.35 / **5.21**	5.87 / **3.83**	3.22 / **1.82 %**
Model P3 (`B1+B2v1 Conv3D`)	2.13	1.25	7.30	5.20	2.61 %
Model P4 (`B1+B2v2 ResNet18`)	**1.02**	**0.71**	5.34	3.95	1.85 %

6 Next Steps for Week 4–5

1. **Gate & Attention Visualisation (W4-06):** Generate per-frame gate traces and cross-modal attention heatmaps from `checkpoints/p_proposed_best.pt` (`P4`).
2. **Ablations X1–X3 (W4-07) & CLI/Gradio Integration (W4-08–10):** Smoke-test X1–X3 configs, wire `transcribe.py`, and upgrade Gradio Mode A to display live fusion gate plots using Model P4.